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Policy Wind Tunnel

Test how people will read a policy before it ships

Live: https://wind-tunnel.stargxzer.me Code: https://github.com/beaaan/seaxopenai-hackathon

In one paragraph. Marketplace rules are enforced as written but acted on as understood. Policy Wind Tunnel gives the exact text of a policy to 50 simulated Vietnamese shoppers who meet it the way real buyers do (help article, checkout message, a rumour from a friend), measures what each one comes to believe, groups the misreadings and traces them to the sentence behind them. A Codex agent decides who reads next. Every proposed edit is fact-checked and re-read by the same readers, and it passes only if the improvement beats the measured test–retest noise: continuous integration for policy text. On Shopee's real COD help article, **16 of 50 readers** came away believing a "three failures in 90 days" rule that Shopee never states; a second run with 50 new readers found the same misreading first (18 of 50). A full run costs about **$2.56** in model calls; re-testing an edit costs about **$0.11** and takes 17 seconds.

1 The problem

Shopee Vietnam's help article 79294 explains why cash on delivery (COD) can disappear from checkout. For the failed-delivery lock it says only that the number of unsuccessful COD deliveries in the buyer's history is

> "*…có số lượng đơn hàng COD giao không thành công* **vượt quá quy định hiện hành**."
> ("…exceeds the current rule.") How many failures? Counted over what period? Locked for how long? The page does not say.

Buyers fill the gap from coupon sites and blogs: "3 failed COD orders within 90 consecutive days locks COD" (magiamgiashopee.vn), "the lock lasts about 60 days" (viettelstore.vn). None of these numbers appears in Shopee's text. The stakes are concrete:

- **77.5%** of Vietnamese e-commerce consumers choose COD [1], and Shopee carried **56%** of 2025 GMV on the four largest marketplaces [2].
- **8–12%** of online orders end in "bom hàng", depending on category and area [3]; returning a failed inter-province delivery costs the shop another **50%** of the shipping fee [4].
- A live support contact costs **$8.01** on average, against about $0.10 for self-service [5].
- Wording measurably changes behaviour: one US Veterans Affairs letter rewrite cut related calls by **more than 80%** [6].

Code is tested before release; policy text is reviewed by people who already know what it means. The existing options are slow or shallow. Paying 50 people for a one-hour interview costs about $4,000 in incentives alone [7]. A single AI review lists possible problems but cannot say *how many* readers believe each one, *who* they are, or *whether a fix works*.

2 What we built

Policy Wind Tunnel is a pipeline, a CLI (`pwt`), an HTTP/SSE server and a web app. The web app has a landing story, a report per run, a replay in which the reader flow (what they heard → where they read → what they believe) fills in as readers report back beside the reviewer agent's tool calls, and a policy lab: pick a ready-made policy or paste one and watch up to 50 readers read and get graded live. A presenter console drives the landing during a talk and puts any of 14 sourced answers on a judges' Q&A screen. Every stage appends to an event log; the UI, the replay and the report are all derived from that log.

- **Readers meet the rule the way buyers do.** Besides traits extracted from the policy (payment options, shared account, delivery history), every reader has an *exposure channel* (help article, checkout message only, nothing official), a *prior rumour* taken from real, sourced folk versions, and an *attention* level. A covering array makes every pair of trait levels appear. Readers are told to be no more careful than their persona and may be wrong.

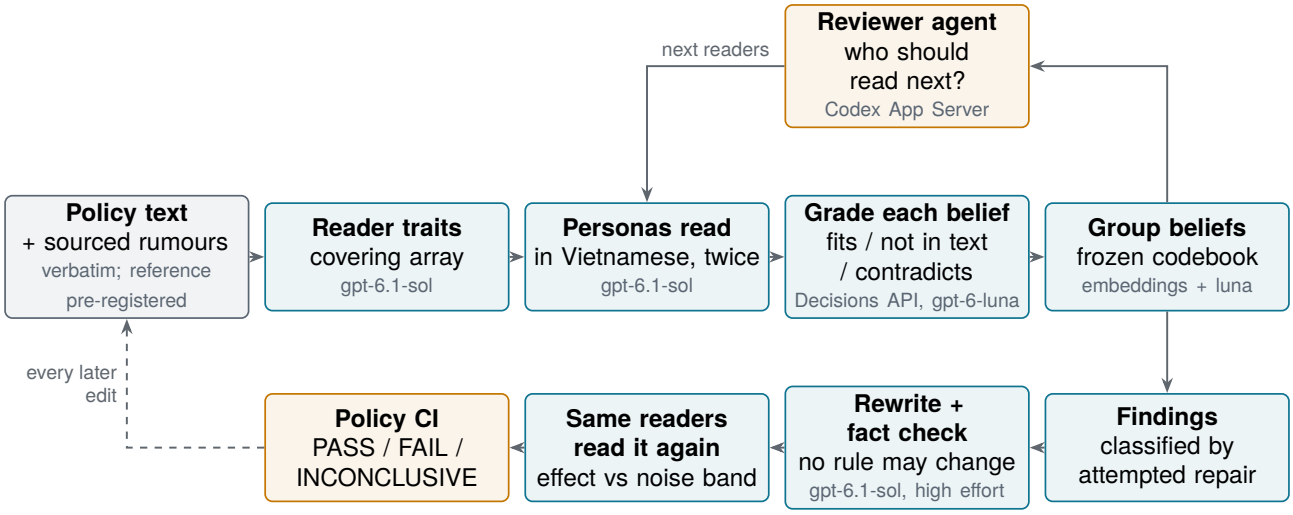


Figure 1: The pipeline. Cyan stages measure; amber stages are agents or decisions.

- **One instrument for before and after.** Each belief is graded against reference facts that were committed before the first run. Beliefs are grouped once into a frozen codebook, and every reading of every version is coded on it (11,880 probability judgements in the main run), so a before/after difference is not an artefact of re-clustering.
- **Noise is measured, not assumed.** Every reader reads twice. An edit counts only if the change exceeds that test–retest band.
- **Findings are classified by trying to fix them.** A minimal rewrite is attempted and fact-checked: *wording fix* if a rewrite keeps every rule and resolves it, *design gap* if any fix must change a rule, *unresolvable* if the text lacks the needed fact.
- **Policy CI.** `pwt test edit.json --suite <run>` re-runs the frozen suite on an edit and fails it if the fidelity check finds a new promise or a dropped fact, or if a misreading gets more common beyond noise.

2.1 What it found (run `cod-main`, 50 readers, 15 minutes)

Headline	**16 of 50** readers believe a three-failure / 90-day cutoff is Shopee policy. 122 first-read beliefs → 47 groups → 7 misreadings, each traced to a sentence.
The official text does not dislodge the rumour	Of 22 readers told "3 failures in 90 days", 16 kept it: 3 of 5 who read the help article, 6 of 7 who saw only checkout, 7 of 10 who saw nothing official. None of the 13 readers told no rumour invented it. 11 of 15 told "60 days" kept expecting a 60-day lock.
The obvious fix fails, and the tunnel catches it	Adding "this article does not state how many failures count, or over what period" kept 10 of 10 facts with no new promise, yet belief in the cutoff moved only 0.85 → 0.82 (inside the ±0.12 band) and another misreading rose from 0.17 to 0.28. Only 3 of the 10 readers ever see the help article. *Next to test: publishing the actual threshold where buyers see it, at checkout.*
A bad edit is rejected	"Pay at a convenience store" for cash-only buyers was rejected: the original text never promised it.
A good edit passes	Grab VN scheduled-ride terms: a rider-clause rewrite lowered the mean belief score of the top misreading from 0.97 to 0.62 on the same 12 readers, beyond the ±0.05 band, with every rule unchanged.

It replicates. A second run with 50 new readers and age group (18–25, 30–45, 55–70) and digital literacy designed in (`cod-demo`, $3.24 metered, 13 minutes) found the same misreading first: **18 of 50**. In this sample, what readers heard (18 of 23 told the rumour vs 0 of 11 told nothing) and where they read (1 of 16 help-article readers vs 9 of 14 who saw nothing official) showed much larger differences than age (9/19, 5/18, 4/13) or literacy (9/22 low vs 9/28 high; largest demographic gap $p = 0.08$).

Who misreads it (holders of the top misreading)

Heard "3 failures in 90 days"	**16/22**
Heard "locked for 60 days"	0/15
Heard no rumour	0/13
Saw nothing official	7/14
Saw only the checkout message	6/18
Read the help article	3/18
Shares the account / own account	11/27 · 5/23
Skimmed / read carefully	10/29 · 6/21

What one misreading can cost (hypothetical)

100,000 buyers locked out × an assumed **10%** who misread × 25% who then contact support (assumed) × $8.01 per contact [5] ≈ **$20,025**. The landing's calculator starts with the misreading rate blank: simulated counts are not population rates, so the viewer supplies all three assumptions. Finding the misreading took one run, estimated at about **$2.56** in model calls.

3 Build directions

Deep Domain AI primary	Built on verbatim Vietnamese marketplace text and the rumours that actually circulate, with sources. Personas are Vietnamese COD shoppers who meet the rule through checkout, the help centre or a friend. The analysis separates rules the text conflates (the 24-hour daily COD cap versus the failed-delivery lock) and ends in a policy decision, not a readability score.
Autonomous & Adaptive AI	The reviewer agent runs the sampling loop on its own: it reads where beliefs split by trait, spends the reader budget there and stops by a logged rule (new beliefs per round $9 \to 5 \to 5 \to 2$). Its 17 tool calls and every rationale are in the event log. Agent threads run isolated (empty working directory, built-in file tools off), so no persona can read the held-out answers.
AI-Native Products & Operations	Policy CI turns a policy team's review into an automated gate: every edit is re-tested on the same readers for cents, with a PASS / FAIL / INCONCLUSIVE verdict and the evidence behind it.

4 How we used LLMs

4.1 Inside the product

All calls go through one gateway (cache, rate limiter, zod validation of every output, optional token log). We matched each API to the shape of the answer: labels and probabilities go to the Decisions API, generated text to Structured Outputs, agent loops judges can watch to Codex App Server.

Role	Model · API	What we ask	Cost*
Readers	gpt-6.1-sol (low) · Structured Outputs	"You are this shopper; a friend told you this; you see this screen. What do you believe now?"	$0.27
Grader	gpt-6-luna · Decisions	Does this belief fit the policy, contradict it, or go beyond it?	$0.02
Grouping	embeddings + sol + luna	Is this belief the same idea as that one?	$0.36
Reviewer agent	gpt-6.1-sol · Codex App Server	Which trait splits beliefs? Who reads next?	$0.10
Findings, repair, fidelity	gpt-6.1-sol (high)	Name the misreading; fix it without changing a rule; did a rule change?	$0.36
Baselines + blind scoring	3× sol, 1× gpt-6-astra	One-shot expert review, scored blind against held-out edge cases	$0.37
Persona agent	gpt-6.1-sol · Codex App Server	Live 180 simulated days under the rule; what do you do?	$0.04

*Measured: a fresh 20-reader run logged every call's tokens (372 calls, $1.54 at OpenAI list prices, 10 minutes). Scaled to 50 readers, a full run is about $2.56.

4.2 Building it with Codex

- **gpt-6-astra as a standing reviewer** (`codex exec -s read-only`): 14 logged reviews. The first, at 10:30 before any code, critiqued the plan and measurement protocol (31 items: paired noise band, frozen codebook, classification by attempted repair, blind holdout scoring). Then module-by-module code reviews, then audits of every number on the landing page and in the judges' Q&A against the logged runs and the cited sources (the last one corrected 14 of 14 flagged statements).
- **Real catches.** App Server persona threads had inherited Codex's shell and file tools, so a persona could have read the held-out edge cases; they are now isolated. Edit re-tests were pinned to the suite's original version and input hashes. Overstated claims were corrected before the demo (for example "more than 80%", not "80%"; belief scores stated as 0.97 → 0.62, not as percentages of people).
- **gpt-6.1-sol task agents** for self-contained slices, run in parallel with file ownership agreed up front: per-stage token metering and the cost probe; sourced research for the stakes section (each figure with quote, date and URL; unverifiable figures were kept out); designed age and literacy traits plus the `cod-demo` run; the replay rebuilt as a reader-flow diagram; the policy lab with ready-made policies and live grading; the presenter console and judges' Q&A screen.
- **Codex at runtime.** The reviewer and the persona agent are Codex App Server threads with our tools exposed as dynamic tools; the replay shows their tool calls next to the reader flow, as they happened.
- Every Codex task is logged with its outcome in `docs/codex-log.md`.

5 Limits

Simulated readers are not a population sample: counts come from a designed mix of traits and are not national rates, which is why the cost calculator takes the misreading rate as an assumption rather than reusing a simulated share. Differences between reader groups, including the age and literacy comparison in `cod-demo`, are observed in a designed, adaptively sampled sample and do not establish effects. The tool's job is to show which sentence to put in front of real people; the app includes a phone page for asking real buyers, coded on the same codebook. The checkout message is quoted as reported by salework.net and has not been verified in the app.

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